

Table 1: CIFAR-10 ResNet-18 layer summary. The CIFAR-adapted stem replaces the standard 7×7 convolution with stride 2 and max-pooling with a 3×3 convolution with stride 1 followed by an **Identity** layer, preserving the full 32×32 spatial resolution through layer1. Total trainable parameters: $\approx 11.17\text{M}$.

Layer (type:depth-idx)	Input Shape	Output Shape	Param #
Conv2d: 1-1	[1, 3, 32, 32]	[1, 64, 32, 32]	1,728
BatchNorm2d: 1-2	[1, 64, 32, 32]	[1, 64, 32, 32]	128
ReLU: 1-3	[1, 64, 32, 32]	[1, 64, 32, 32]	--
Identity: 1-4	[1, 64, 32, 32]	[1, 64, 32, 32]	--
— layer1 (Sequential: 1-5)			
BasicBlock: 2-1	[1, 64, 32, 32]	[1, 64, 32, 32]	--
Conv2d: 3-1	[1, 64, 32, 32]	[1, 64, 32, 32]	36,864
BatchNorm2d	[1, 64, 32, 32]	[1, 64, 32, 32]	128
Conv2d: 3-4	[1, 64, 32, 32]	[1, 64, 32, 32]	36,864
BasicBlock: 2-2	[1, 64, 32, 32]	[1, 64, 32, 32]	73,856
— layer2 (Sequential: 1-6)			
BasicBlock: 2-3	[1, 64, 32, 32]	[1, 128, 16, 16]	230,144
BasicBlock: 2-4	[1, 128, 16, 16]	[1, 128, 16, 16]	295,424
— layer3 (Sequential: 1-7)			
BasicBlock: 2-5	[1, 128, 16, 16]	[1, 256, 8, 8]	919,040
BasicBlock: 2-6	[1, 256, 8, 8]	[1, 256, 8, 8]	1,180,672
— layer4 (Sequential: 1-8)			
BasicBlock: 2-7	[1, 256, 8, 8]	[1, 512, 4, 4]	3,670,016
BasicBlock: 2-8	[1, 512, 4, 4]	[1, 512, 4, 4]	4,720,640
AdaptiveAvgPool2d	[1, 512, 4, 4]	[1, 512, 1, 1]	--
Linear (fc): 1-10	[1, 512]	[1, 10]	5,130